

CURRITUCK COUNTY REGIONAL, NC, USA

WMO: 723626

Lat: 36.399N

Lon: 76.016W

Elev: 16

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.5	25.9	2.9	6.4	28.7	8.5	8.5	31.6	20.9	43.0	19.5	40.0	5.3	320	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	92.6	76.6	90.4	75.8	88.3	74.8	79.4	87.5	78.2	85.7	77.3	84.3	7.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	142.5	82.5	76.4	137.7	82.0	75.1	131.8	81.2	42.9	87.2	41.7	85.3	40.6	84.6	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	17.1	15.2	DB	16.0	97.1	6.1	2.0	11.6	98.6	8.1	99.8	4.7	101.0	0.3	102.4	(4)
(5)			WB	13.8	81.1	5.4	2.0	10.0	82.5	6.8	83.7	3.8	84.9	-0.1	86.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	42.3	44.5	50.1	60.3	67.8	75.9	78.9	78.4	72.5	63.2	52.4	47.0
	DBStd	15.21	10.25	9.49	9.61	8.59	7.46	5.59	4.49	3.84	5.29	8.37	8.51	9.33
	HDD50	872	281	207	124	12	0	0	0	0	0	8	69	171
	HDD65	3094	703	576	470	193	60	2	1	0	6	136	385	562
	CDD50	4958	43	52	127	321	551	778	896	880	675	416	141	78
	CDD65	1704	0	1	8	51	146	330	432	415	231	79	7	4
	CDH74	13975	0	7	57	361	1050	2855	4136	3656	1437	389	19	8
	CDH80	4545	0	0	5	77	300	1033	1522	1209	332	66	1	0
(14) Wind	WSAvg	5.9	6.4	6.9	7.6	7.6	6.2	5.4	4.5	4.3	5.6	5.4	5.7	5.7
(15) Precipitation	PrecAvg	48.7	3.7	3.0	3.8	3.3	3.8	4.6	5.7	5.5	4.9	3.9	3.1	3.5
	PrecMax	65.4	6.9	6.5	8.1	7.2	7.7	10.7	12.6	9.8	12.7	9.7	8.3	8.6
	PrecMin	29.9	1.1	0.8	1.2	1.0	0.9	1.2	1.4	1.9	0.8	0.1	0.2	1.1
	PrecStd	8.4	1.6	1.4	1.6	1.5	1.7	2.0	2.2	2.2	3.2	2.3	2.0	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.4	74.4	80.1	86.3	91.7	95.5	95.2	94.6	90.1	85.4	76.7	74.7
		MCWB	62.6	63.0	62.8	68.0	71.5	74.9	78.1	78.1	74.6	72.3	67.4	67.2
	2%	DB	66.3	69.5	74.7	82.1	87.1	91.5	92.6	90.9	87.1	81.6	72.4	69.9
		MCWB	60.8	59.3	61.3	66.0	71.3	75.4	77.5	77.2	73.5	70.7	64.6	65.1
	5%	DB	63.1	65.0	70.4	78.6	83.5	89.2	90.3	88.9	83.7	78.5	69.4	65.9
		MCWB	57.6	57.8	59.3	64.5	70.2	73.9	76.6	76.1	72.8	69.5	62.5	62.2
	10%	DB	59.1	60.8	66.0	74.2	80.0	86.1	88.0	86.3	81.2	75.1	66.0	62.8
		MCWB	54.5	54.5	58.6	63.2	68.0	73.2	75.7	75.1	72.2	68.7	60.9	59.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	64.8	66.7	73.1	75.3	78.5	81.1	81.1	77.4	76.4	70.7	69.9
		MCDB	67.5	71.1	74.9	78.8	84.2	87.2	90.1	89.5	83.8	81.0	73.1	72.0
	2%	WB	61.9	62.3	64.3	69.7	73.5	77.4	79.6	79.6	76.2	74.4	67.1	66.0
		MCDB	65.6	67.4	70.8	77.7	82.7	86.0	87.8	87.5	82.1	78.2	70.3	68.7
	5%	WB	58.8	59.5	62.1	67.3	71.9	76.3	78.3	78.3	75.2	72.6	64.3	63.0
		MCDB	62.2	64.0	67.8	74.4	79.4	84.6	86.2	85.3	80.7	76.3	67.7	65.3
	10%	WB	55.1	54.8	59.4	65.1	70.5	75.1	77.3	77.3	74.2	70.2	61.8	59.4
		MCDB	58.8	59.9	65.6	72.3	77.4	83.0	84.8	83.5	79.4	73.9	65.1	62.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	16.8	16.9	17.6	16.8	15.6	14.9	14.0	14.0	16.3	17.5	16.3
		MCDBR	22.0	22.5	22.9	22.0	20.1	18.9	17.4	16.8	17.2	17.8	19.2	18.3
		MCWBR	16.9	15.0	12.6	9.8	8.0	7.0	6.4	6.0	7.2	8.3	12.4	14.3
	5% WB	MCDBR	20.7	21.4	19.6	18.4	17.1	16.6	15.8	14.9	13.5	13.1	16.4	17.4
		MCWBR	18.1	16.6	13.6	10.7	7.8	6.9	6.4	6.0	6.3	6.9	13.4	15.2
(40) Clear-Sky Solar Irradiance	taub		0.314	0.325	0.351	0.394	0.438	0.487	0.514	0.504	0.412	0.361	0.329	0.319
	taud		2.535	2.497	2.454	2.344	2.254	2.168	2.104	2.121	2.376	2.505	2.542	2.550
	Ebn at Noon		281	290	291	282	270	256	247	247	267	273	272	271
	Edh at Noon		25	29	33	39	43	47	50	49	36	29	25	23
(44) All-Sky Solar Radiation	RadAvg		750	994	1331	1702	1912	2031	1931	1742	1432	1144	847	673
	RadStd		59	102	101	121	147	121	95	110	103	123	64	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+202

Nomenclature: See separate page